

Data Warehouses

Multidimensional model and relational storage

- What is the multidimensional model?
- Can you please give an example for it?
- Please, explain the facts in Data Warehouses, the characteristics of the facts, dimensions, and queries
- How can this model be translated into relational databases?
- For an example: how does a Star schema looks like?
- For the same example: how can you store multidimensional data in a Snowflake schema?
- What are the differences between Star and Snowflake schema?
- What kind of normal form do we achieve if we transform a Star schema into a Snowflake schema?
- What does fullfact schema means?
- Why is this storage method only seldom applied in data warehouses?

SQL extensions

- Grouping sets, rollup, cube (syntax)
- Explanation of the keywords
- For all: sample query (for a given star schema)

Multidimensional storage

- General Idea: how is a multidimensional model translated into a linear representation?
- Why is the order of dimensions relevant for efficiency of multidimensional storage?
- How can we find a dataset within the linear representation of the multidimensional data model?

MDX

- Only general idea
- Why MDX has been developed? What is the reason for the development of a new query language for Data Warehouses?
- Which operations are available? (selection, filter, crossjoin)
- *Hint: the concrete syntax of the MDX queries is not relevant for the exam*

Column Stores:

- General idea how to realize column stores?
- Which kind of compressions can be applied?

- Can you please explain each of the compression method?
- In which cases, which kind of compression delivers good results and should be applied?
- Why column stores are mainly be applied for Data warehouses?
- Can you please explain the idea of invisible join?
- What are the differences between early and late materialization?
- What are the advantages of late materialization?

ETL Process

- Extraction:
 - o How can the extraction of relevant data be organized? (time based, event-based)
 - o Can you please explain the differential snapshot algorithm?
 - o Usage of timestamps in this alg.
- Transformation:
 - o What are the differences between Schema and Data Mapping?
 - o Why is error recognition easier to program then error correction?
 - o Why do we use a Staging Area for it?
- Load:
 - o What are the differences between bulk load and tuple based load operations?
- Different data sources in the ETL process (NoSQL databases)
 - o Which kinds of NoSQL databases exist?
 - o Which are the underlying data models?
 - o What are the differences between NoSQL and SQL databases?
 - o What do you have to consider during loading data from schema less databases into the data warehouse?
 - o Which kinds of structure information can be extracted from available NoSQL databases/XML documents

Best practices in Data Warehousing

- Imagine, you are responsible for a Data Warehouse project: what would you recommend? Can you please explain some of the best practices in data warehousing?

Data Mining

Association Rules

- What is an association rule? With which two measures can you define the reliability of an association rules?
- How does the apriori approach works?
 - (algorithm, apriori step, complexity, how many table scans)
- Could you explain the fp growth algorithm?
 - (algorithm, how many table scans)
- How can the confidence be calculated?
- Advantages of fp growth in contrast to the apriori algorithm
- Application fields for association rules

Clustering

- Can you please define, what is a cluster?
- How can you determine the similarity of two objects?
- Handling of null values?
- How can we determine similarity between two attributes with string types?
- K-means algorithm
 - Basic idea of the Algorithm
 - Can you explain the algorithm with an example?
 - Can you sketch the algorithm (in pseudo code).
 - In which cases, it is complicated to find a center point of the clusters, how can we solve the problem?
 - Complexity of the algorithm
- Hierarchical clustering
 - Can you explain the general idea of an hierarchical clustering algorithm (bottom-up)
 - How long do we execute the algorithm (which conditions can be used for terminating the algorithm)
 - Why is it meaningful to store a similarity matrix during for the execution of the algorithm.
 - Advantages of the hierarchical clustering algorithm/ disadvantages
 - Complexity
- Density-based algorithm
 - (only general idea: input information, number of objects, radius, ..)
 - Which kinds of clusters are created

- Comparison of the different clustering algorithms

Classification

- How can we derive a decision tree from a set of objects with a classification attribute? (concrete calculation of entropy and gini index not necessary, but the characteristics)
- Divide- and- conquer: Why is it a divide-and-conquer approach? When does the process stops?
- What are the differences between training set and test set?
- How can we prove/calculate the quality of a decision tree?
- What does cross validation mean?
- Derivation of rules from decision trees

Case-based reasoning

- Main idea of the subtasks retrieve, reuse, revise, retain
- Retrieve: how can you find similar cases?
- Retain: how can a case bas be organized?